

GDD

≡ Description	The single source of truth of UASU - Game Design Document
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Game Design Document

18/01/2026

Sofía Cabrera
Edurne Delgado
Maxence Dupuis
Leyre Gallud

UASU

A CASCA STUDIO GAME



FACULTAT DE BELLES ARTS DE SANT CARLES

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Maxence Dupuis 🧑

Eduarne Delgado

Sofía Cabrera

Leyre Gallud

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1. INTRODUCTION

1.1 SCOPE OF THE DOCUMENT

This document is primarily intended for all current members of the team behind the game's development. The current team is composed of:

- Sofía
- Edu
- Leyre
- Max

Secondly, it is addressed to anyone interested in the design and development of the game, as well as potential new members who wish to contribute to the project's growth. Finally, it may also serve as a reference for potential partners or investors interested in supporting a larger-scale version of the project.

1.2 GANTT DIAGRAM

UASU

Desarrollo de Videojuego - 2025/2026

Casca Studio

Gantt Chart



Figure 1: Gantt Chart for UASU project development timeline.

1.3 ELEVATOR PITCH

1.2.1 MAIN IDEA

A game where the player takes on the role of Uasu, the main character. Uasu's people, known as the Sifflers, have been imprisoned on a hostile island. One day, Uasu awakens on this island with a single

goal: to free their people. The experience focuses on exploration, puzzles, and combat, with strong emphasis on lore and storytelling to deeply engage the player in the world.

1.2.2 PLATFORMS

The game is developed using Unity 3D Engine v6 and initially designed for PC with keyboard & controller support (PlayStation 5 DualSense controller used during development).

1.2.3 SWOT ANALYSIS

We conducted a member-based SWOT to better understand each team member's strengths and abilities, as well as a game-oriented SWOT to analyze the project's overall potential and challenges.

1.2.4 SOLID WORKFLOW

We follow an Agile workflow based on the Scrum method to maintain flexibility and communication. Weekly checkpoints are organized to ensure consistent progress and team coordination. Project management is handled through Notion.

- Pre-production: 2 weeks (Sprint 1)
- Production: 8–10 weeks (Sprints 2–5)
- Testing & Launch: 1–2 weeks (Sprint 6)

1.4 TEAM ROLES

Maxence Dupuis 🧑

I handle the coding (git/C#) and 3D (Blender) parts.

If you have a doubt into more advanced coding concepts or 3D workflow questions, don't hesitate to ask!

Edurne Delgado

I'm one of the members of the art department.

I made concept art for characters and modeled 3D and put color to characters and assets. Also designed and made interface on main menu, options, pause menu and HUD. Worked on environment (asset placement) and applied the music. Wrote the "lore" of the game and made the intro cinematic.

Sofía Cabrera

I'm part of the art department.

I created the concept art for the map and for some characters, modeled and colored assets and a character. Also hand-painted the characters. Worked on the environment of each level in unity and chose the music.

Leyre Gallud

I worked on 3D modeling and a bit of coding. I've 3D modeled and applied textures for levels 1 and 2 and the lighting on the Sanctuary. I've also been working on an inventory system for the game.

2. GAME OVERVIEW

2.1 CONCEPT

The goal of the game is to immerse the player in a fictional, cohesive, and visually appealing world. The player will mainly:

- Interact with NPCs and objects (learn more about the story, get hints, improve their skills).
- Technical combat against enemies and boss with a featuring lock on and dodge mechanics.
- Platforming and exploration-based progression.
- Puzzles and environmental interactions.

We want the player to truly enjoy controlling their character and seeing them evolve throughout the game. We aim to give a sense of progression and autonomy, without directing the player too explicitly. The enjoyment will come from their ability to observe and listen to the various elements of the game.

2.2 TARGET AUDIENCE

- Independent players, driven by the will to explore every corner, uncover every hint, and engage in every dialogue of the game (familiar with **Metroidvania** mechanics)
- Fans of **Soulslike** challenges.- Nintendo enthusiasts (**Zelda: BOTW**, **Mario**, etc.)
- Likes original and mysterious art style.

2.3 GENRE

A 3D Action-adventure game.

- > Isometric Camera with 3D models Action
- > Combat Adventure
- > Puzzle, Platforming

2.4 SETTING

It took place in a Fantasy World with a duality between Light & Darkness. A mix of Zelda's world with Dark Souls.

2.5 WORLD STRUCTURE

The player navigates a world that is designed to be linear, but gives a sense of freedom by allowing them to return to previously explored areas. This is also necessary for collecting items that may enable access to areas that were previously blocked, or for interacting with NPCs depending on their progress in the adventure.

There are **4** different levels in the Game.

The difficulty increases with the player progression.

2.6 PLAYER

It's a **single** player game. Player embodies Uasu (the main character).

2.7 CORE LOOP

- Idling
- Walking
- Jumping

- Interaction (NPC's Dialogue, Objects, enemy lock)
- Attack (Even if at the beginning, the player doesn't have a weapon)

2.8 PEGI CLASSIFICATION

The game is rated **PEGI 7** due to the presence of non-realistic, stylized violence. Combat is fantasy-based, with no blood, visible injuries.

The visual style is highly stylized and symbolic, similar to games such as *Tunic* or *Hollow Knight* (both PEGI 7). Violence is conveyed through abstract animations and visual effects, without shocking impact.

The game contains no inappropriate content, including no strong language, no sexual references, no depiction of drugs, alcohol or tobacco, and no discriminatory messages.

Although the world is built around a Light and Darkness duality, the atmosphere remains accessible and poetic, with no horror elements or frightening scenes.

Gameplay focuses on exploration, puzzles, and NPC interactions, encouraging observation and reflection rather than violent behavior.

3. GAMEPLAY

3.1 OBJECTIVES

GAMEPLAY SCHEMA

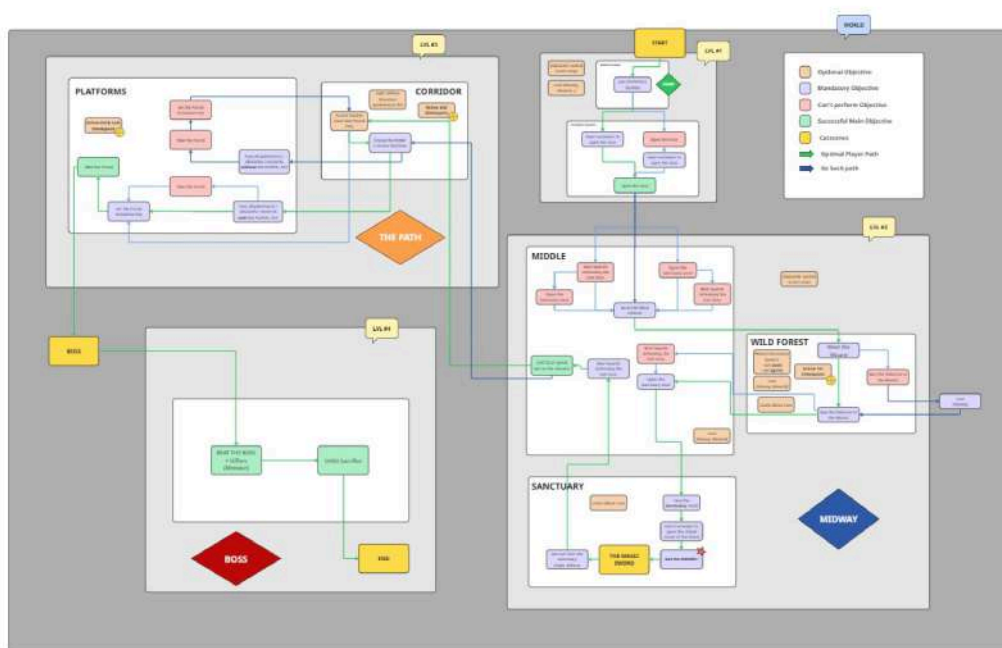


Figure 2: Gameplay Schema for UASU.

3.2 DIFFICULTY

There are currently no "difficulty levels" such as Easy, Normal, or Hard.

We're inspired by *Souls-like* games. Of course, this isn't meant to be an extremely difficult game at this stage, though the challenge is definitely present!

As shown in the previous diagram, some objectives cannot be completed at a given moment unless other objectives have been achieved first. We also want to expose the player to this concept, to "show the problem before the solution." This way, when a zone is blocked, it naturally guides the player, without explicit instructions, to reorient themselves in the right direction.

4. MECHANICS

4.1 RULES

The player must progress through the levels, solve puzzles, and overcome challenges. There are a total of 4 levels to complete. The player cannot go beyond the boundaries of the map. If they attempt to leave the map, we'll let them fall and respawn.

4.2 PHYSICS

Our game physics will be semi-realistic/cartoonish, the environment won't be destroyable if it's not necessary for puzzle solving, etc. We use the latest Unity Input System with the Character Controller component for the player.

4.2.1 ROCK PUSHING SYSTEM

Player can interact with a special rock with a symbol inside the first level of the game. During this interaction, the player needs to push the rock on a pressure plate with the same symbol of the rock. This way, the player can open a door and access the second level.



Figure 3: Interactive rock with the same symbol as the pressure plate.

4.2.2 ROCK FALLING SYSTEM

Inside the Sanctuary zone, when the player retrieves the sword, a scene reminiscent of Indiana Jones or Uncharted unfolds: the Sanctuary begins to collapse, with rocks falling and dealing damage to both enemies and the player. To simulate the physics of this event, we use a Rigidbody. The player can see where the rocks are falling, allowing them to attempt to dodge.



Figure 4: Rock Falling System: Screenshot of the playground scene for testing and developing the rock falling system.

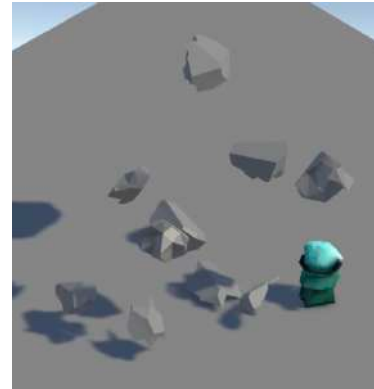


Figure 4.1 : Rock Falling System: Screenshot of the playground scene for testing and developing the rock falling system.

4.3 ECONOMY

The game doesn't need or have a monetary system, but the player will have an inventory in which they will be able to save resources and objects they find in the map to trade with the Sorcerer. This mechanic is important to the story's development, since the player will have to trade to acquire an important item, the artifact, which will allow us to get the magic sword.

4.4 CHARACTER MOVEMENT

Since the game will be developed in an isometric perspective, the player will be able to move in all directions (up, down, left, right and diagonally).

To move the character, we use the Unity Character Controller component and latest Unity Inputs System.

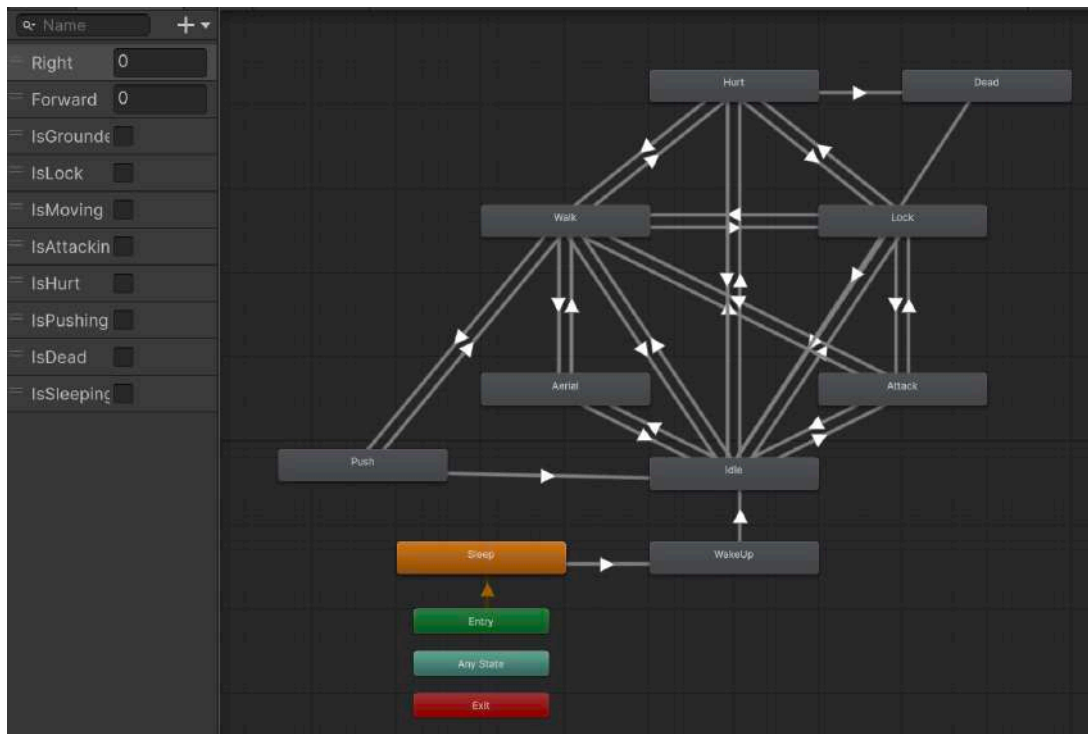


Figure 5: Uasu's Animation State Machine.

4.5 PLAYER INTERACTION

The player will be able to interact with objects, doors, pressure plates, levers, etc... As well as with objects and enemies. Interaction is important in our game, most map elements will not only be interactive but will have a specific use and meaning to the story's progression (Ex. In level 2 the player needs to interact with the Sorcerer to get the artifact).

4.5.1. SAVING

IMPORTANT : we simplify this interaction for the moment because we're not targeting the "data-save" system enabling player to exit the game, come back and continue.

If the player wants to save their game, they must do so via checkpoints, which will save automatically (a small icon will appear). Checkpoints also restore full HP.

When charging an ongoing game, the player will always spawn in the last saved checkpoint.

4.5.2. GAME OPTIONS

The player will be able to change and adapt the different settings of the game options via the Start or Pause Menu.



Figure 6: UASU's start menu.

4.5.3. DIALOGUE SYSTEM

You can interact with some character to start a conversation with them and to unlock certain tasks. This system can handle multiple NPC, and is flexible to targets specific dialogues with precise requirements.

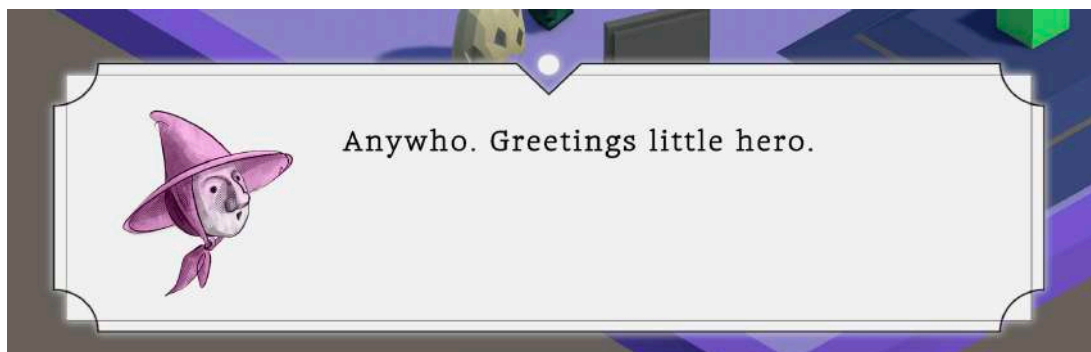


Figure 7: Screenshot of the playground scene to test the dialogue system, the dialogue "card" pops up after the player interacts with the character.

If you want to create new dialogues for a specific NPC, all you need to do is :

1. Going into the json files and add some dialogues with specific ID's (text, next dialogue etc...).
2. Create a "NPC" DialogueTrigger class with all achievement to access specific dialogue.
3. Call the TriggerDialogue() method in the NPC DialogueSate.

```

public class SorcererDialogueTrigger : DialogueTrigger
{
    2 references
    protected override int GetCurrentDialogueID()
    {
        if (PlayerManager.Instance.HasAchievement("WHAT_IS_CALLING_ME_1"))
        {
            return 69;
        }
        if (PlayerManager.Instance.HasAchievement("THE_SORCERER_FLOWER"))
        {
            return 49;
        }
        if (PlayerManager.Instance.HasAchievement("GOOD_LUCK_LITTLE_HERO"))
        {
            return 48;
        }
        return 0;
    }
}

```

Figure 8: NPC Dialogue trigger code: C# script for managing NPC dialogues based on player achievements in a game.

```

{
    "id": 48,
    "speaker": "sorcerer",
    "text": "Now, good luck, little hero!",
    "alternative_text": "Still waiting, and so are they. Go on, hero.",
    "next_dialogue": 49,
    "new_achievements": [
        "GOOD_LUCK_LITTLE_HERO"
    ]
},
{
    "id": 49,
    "speaker": "sorcerer",
    "required_achievements": [
        "THE_SORCERER_FLOWER"
    ],
    "text": "Welcome back hero.",
    "next_dialogue": 50
},

```

Figure 8.1: NPC Dialogue trigger code: C# script for managing NPC dialogues based on player achievements in a game.

```

public class SorcererDialogueState : SorcererState
{
    1 reference
    public SorcererDialogueState(SorcererManager sorcererManager) : base(sorcererManager) { }

    3 references
    public override void Enter()
    {
        UIManager.Instance.Show(PanelType.Dialogue);
        _sorcererManager.DialogueTrigger.TriggerDialogue();
    }
}

```

Figure 8.2: NPC Dialogue trigger code: C# script for managing NPC dialogues based on player achievements in a game.

4.5.4. SOUL GATHERING SYSTEM

Every time you kill an enemy, you collect its soul. A counter on the top right corner appears everytime you gather one, and it shows how many you got. Souls are necessary to progress in the game since, if you do not have the amount of souls it tells you to gather in each level, the game will not let you go into the next one.

In the games story, by “killing” the enemies with the magic sword you are liberating their souls from their evil and dark forms and keeping them safe inside the Artifact. That way, at the end of the game, you will free them all after killing the boss.

4.5.5. HEALTH SYSTEM



Figure 9: Screenshot of the playground scene to test the health system.

While fighting enemies, the player loses HP if hit. If the lifebar empties, it's game over. HP can be fully recovered in checkpoints and also, while in a fight, with a “health item”.

The health item is a fruit that can be found in various spots of the map: inside barrels, on the floor, etc. The player can pick them up and keep them in the inventory to use them while fighting or when there aren't any checkpoints near the area. They recover some of the HP. When the player exits an area of the map, the health items reappear.

4.5.6. BRIDGE SYSTEM

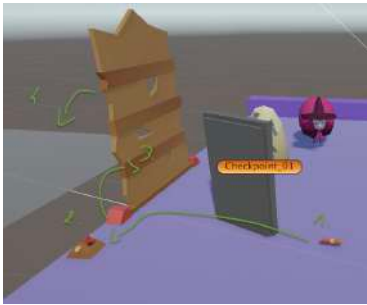


Figure 10: Screenshot of the playground scene to test the bridge system.

Some parts of the map may be inaccessible due to large gaps where players can fall and die if they attempt to jump from one side to the other. One puzzle involves activating a bridge using a mechanism. When activated, the bridge rotates and opens a new path for the player.

To solve this puzzle, players must first find a 'stick' that resembles a lever. They then need to insert the stick into a lever support and strike the support to activate the bridge.

4.6 COMBAT

A big feature is the combat system with a lock-on mechanism. Inspired by souls-like game, this feature enable the player to get more control during a fight. This increase a lot the dynamism.



Figure 11: Yellow Indicator = Not locked (free movement)



Figure 11.1: Red Indicator = Locked

4.7 KEY OBJECTS GATHERING

To get the Artifact, you will need to complete the Sorcerers mission, which consists on looking for three especific objects gathered around level 2. This objects are a flower, a feather and a broken mirror, and you will have to collect them in your inventory and bring them back to the Sorcerer.

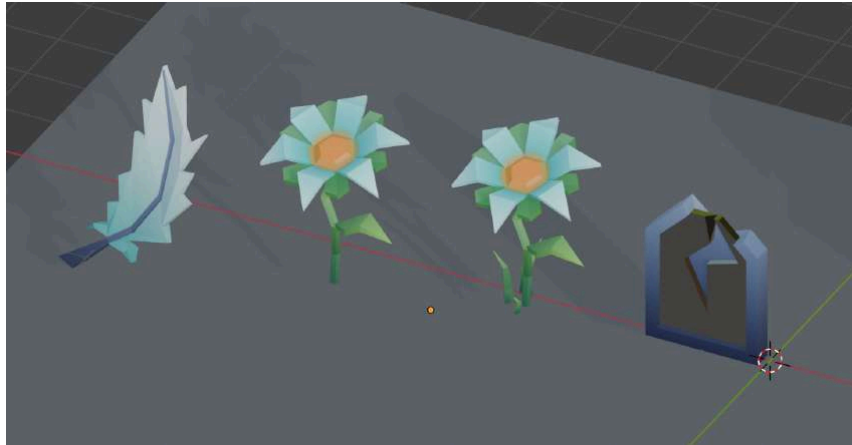


Figure 12: Screenshot of the objects needed to get the Artifact.

Symbolically, these 3 objects represent Uasus state at the beginning of the game: vulnerable, weak and broken. But, when you bring these objects, the symbol of your weakness, to the Sorcerer, he will grant you the artifact, which using would bring Uasu lots of strength.

5. GRAPHICS & AUDIO

5.1 VISUAL SYSTEM

The visual system aims to make the game world feel like a small 3D diorama. Even though the original idea was to make a toy-like environment, given the limited conditions and material, we'll give the game a low-poly style overall. Since it's clean and simple, our scenes won't be saturated.

The landscapes are not overloaded or overstimulating for the player; exploration should not be frustrating or tiresome. Rather, it should be fluid and entertaining.

The game's color moves away from traditional palettes like Nintendo's, green grass and trees or blue sky. The overall color palette in the game is made up of tones that follow analogous or complementary color schemes. Prioritizing arbitrary tones to build specific atmospheres.

In the demo level, as the player approaches the final phase, a saturation gradient can be seen in the environments, starting with warm and comforting tones and reaching a cold and gloomy area.

5.1.1 PLAYER CAMERA

The game's camera is fixed isometric, following the main character, showing a large portion of the map. A 15-20% zoom is implemented when:

- The player approaches or interacts with an NPC.
- The player passes through an important area.
- The player approaches an object requiring interaction (material, checkpoint, etc.).
- Dialogue.

Tilt-shift blur provides a depth that is very difficult to achieve with this type of framing.

5.2 MAP

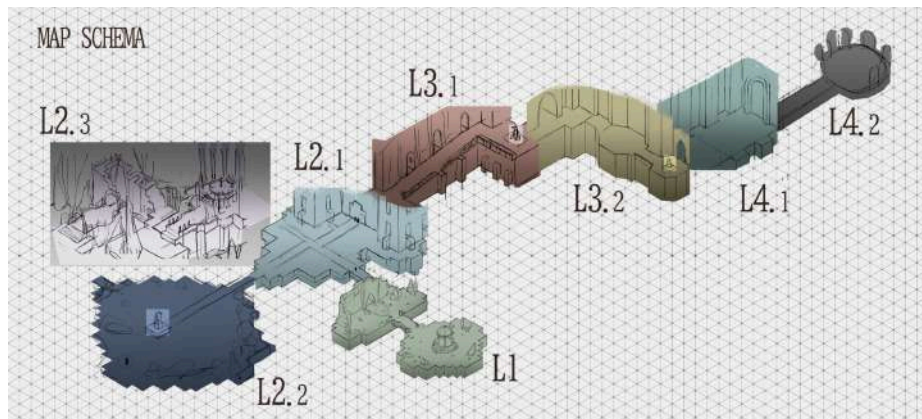


Figure 13: UASU's map schema.

The game takes place on floating land platforms, allowing us to see the sky with clouds in the background. The game demo is divided into four levels, each with its corresponding setting:

- **LEVEL 1: Outside the Walls**

Two platforms, one circular and one rectangular, connected by a drawbridge. This is the exterior area of the castle/fortress. We find ourselves on a grassland terrain, with plenty of vegetation but not high enough. In the middle of the first platform, we find a circular lake under a ruined gazebo that blends seamlessly into the natural landscape. The second platform displays the same flora as the circular platform, but features a high stone wall with a large closed gate. Warm tones predominate.

- **LEVEL 2: Outside Courtyard**

Outside area, now within the castle grounds. 3 sub-areas::

- **Central:** Outside area. High walls of more polished stone protected by two guards. A large gate gives access to the castle. The terrain is sparsely vegetated, more earthy. Decorated with some battle equipment: shields, spears, etc. Warm tones predominate.
- **East:** Outdoor area with more vegetation, bordered by a forest. Here we find a small traveling stall belonging to the mage who will assist the protagonist. We also find the first save point, a statue (larger than the character) ancestral to the sifflers. Clicking on it generates a small draft that makes the characteristic whistling sounds of this race. In addition to the stall and the checkpoint, there is also a crafting table that the protagonist must return to after level 3 to be able to craft the solution they must apply to the sword to defeat the final boss. This level also features various materials that the protagonist can/must collect to advance the main quest. Warm tones predominate.
- **Forest clear:** In a small path of grass surrounded by trees you will find the Sorcerer, a weird individual that will assign you a mission as exchange for an important item: the Artifact. He must get 3 different items from the east zone, and bring them back to him.
- **Sanctuary (north):** Interior area. Crossing a small clearing hidden in the forest barrier in the eastern area, you reach the sanctuary area. A small underground cave hides the sword needed to access the castle. Two stone platforms are connected by a narrow path, and in the background, a structure with an arch and columns surrounding the weapon the protagonist must collect. All surrounded by water. Predominantly, cool tones.

- **LEVEL 3: Inside the fortress (NO IMPLEMENTED)**

Interior area. Divided into two platforms.

The first is a very long, sloping corridor inside the castle. The castle is made of stone, medieval in style, but it is not at all ostentatious; on the contrary, it is decadent and neglected. Nature is beginning to make its way inside. On both sides of the slope, there are small "rest" areas so the player can complete the level. In one of these, the key needed to advance to the next level will be hidden. At the end of the hallway, there's a checkpoint located before the passage to the second platform, another Siffler statue.

The second platform is an elongated rectangular platform, where numerous enemies will appear. At the end, there's the portal that leads to the boss. This is where the third checkpoint is located.

This area is filled with rounded arches, solid stone, and torches. The lighting relies on the torches, so the light will be dim and warm, although the tones become more desaturated as you move towards the end.

- **LEVEL 4: Boss Lair Interior area. (NOT IMPLEMENTED)**

Dark room with a directional light in the center that allows you to see a semicircle of metal sculptures/armors that represent the Sifflers' past. The tone is ancient, stagnant, and even a little melancholic. Predominance of desaturated tones.

5.2.3 FINAL CONCEPT



Figure 14: Final concept art for Level 1: Outside the Walls.



Figure 15: Final concept art for Level 2: Outside Courtyard. Central.



Figure 15.1: Final concept art Level 2: Outside Courtyard. East



Figure 15.2: Final concept art for Level 2: Outside Courtyard. Forest Clear.



Figure 16: Final concept art for the Sanctuary: Secret Cave.

5.2.4 FINAL 3D



Figure 17: Final 3D model of Level 1: Outside the Walls.

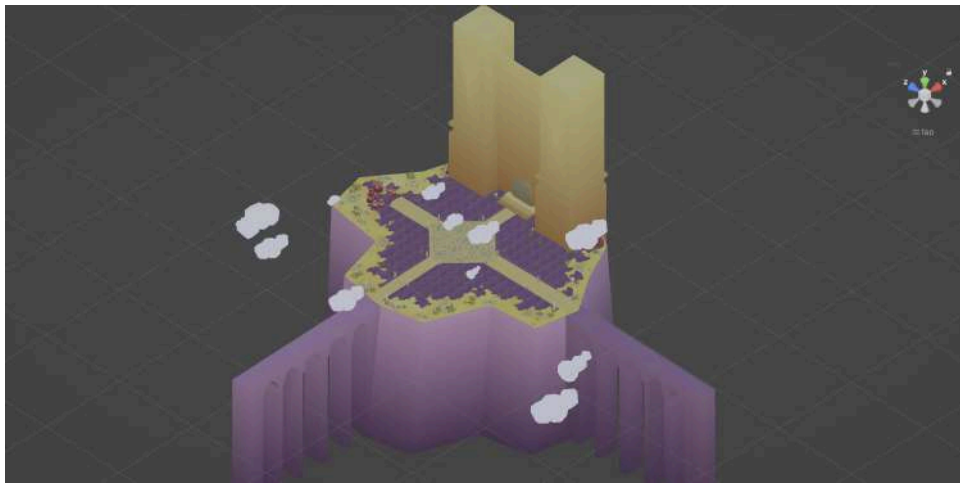


Figure 18: Final 3D model of Level 2: Outside Courtyard. Central.

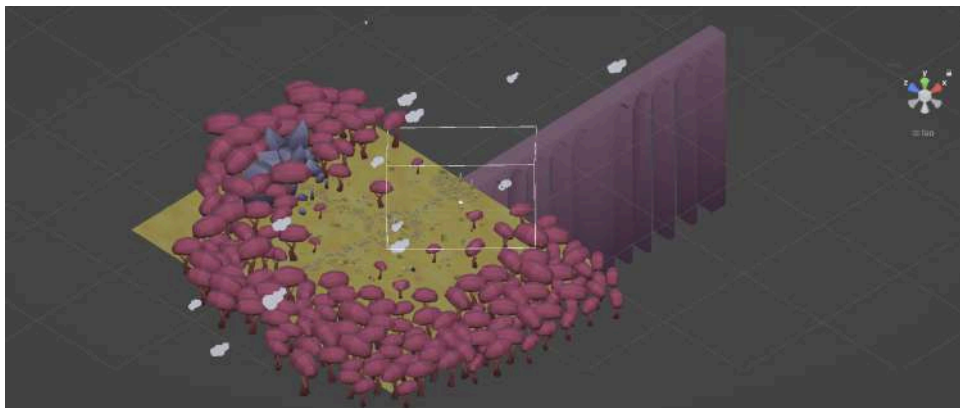


Figure 18.1: Final 3D model of Level 2: Outside Courtyard. East

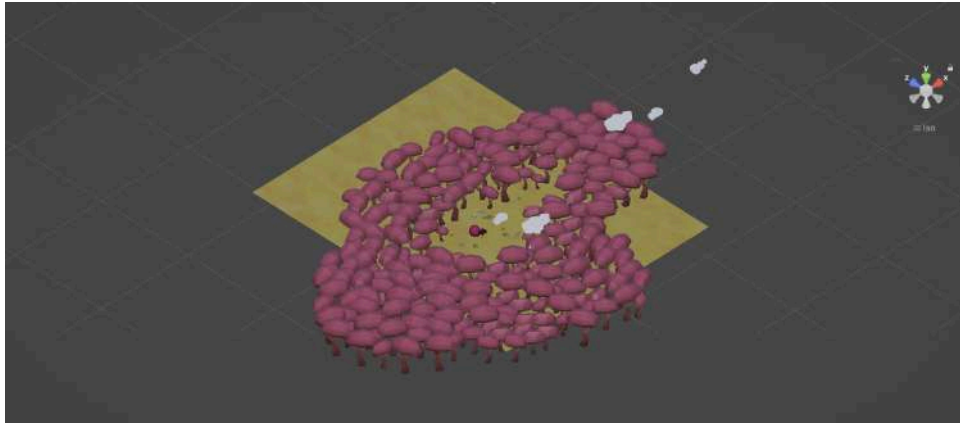


Figure 18.2: Final 3D model of Level 2: Outside Courtyard. Forest Clear.

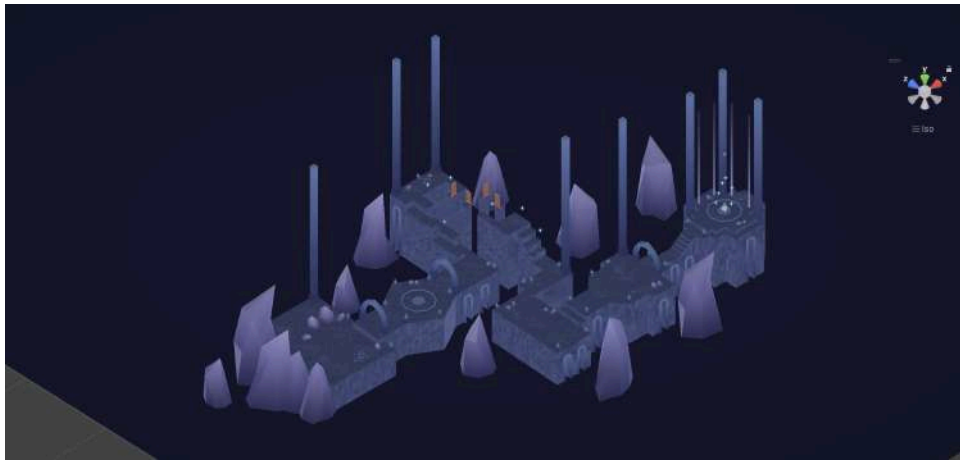


Figure 19: Final 3D model of the Sanctuary: Secret Cave.

5.2.4 TEXTURES

Las texturas se aplicarán sobre todo a los respectivos suelos de cada nivel del mapa:

- Level 01:

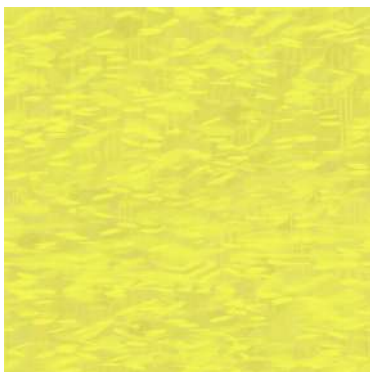


Figure 20: Level 1 floor pattern.



Figure 20.1: Level 1 floor pattern border.

- Level 02:

- Sanctuary:



Figure 21: Level 2 floor pattern.



Figure 22: Sanctuary floor pattern.

5.3 PROPS

5.3.1. MODELING

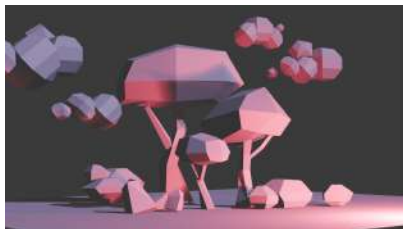


Figure 23: Final 3D model of the trees, clouds, bushes and rocks used in the environment.



Figure 26: Final 3D model of the Sanctuary pivots.



Figure 28: Final 3D model of the grass.



Figure 24: Final 3D model of the columns for Level 2.



Figure 27: Final 3D model of the Level 2 barrels.

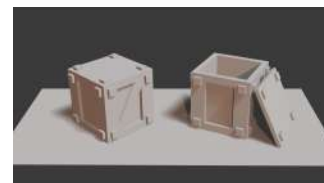


Figure 29: Final 3D model of the boxes for Level 2.

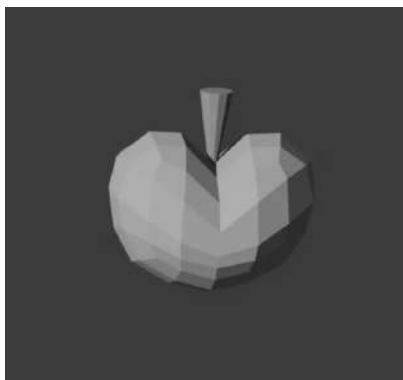


Figure 25: Final 3D model of the Health Item.

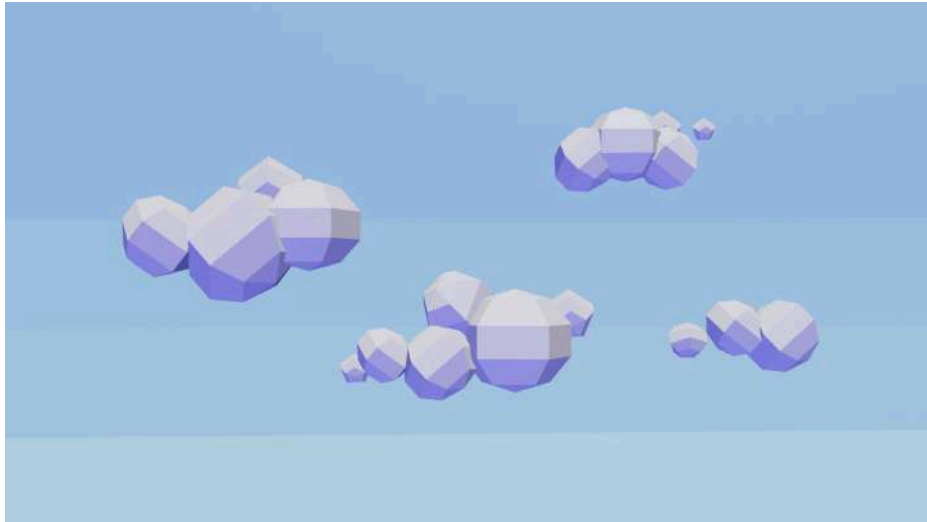


Figure 30: Final 3D model of the clouds in Level 1.

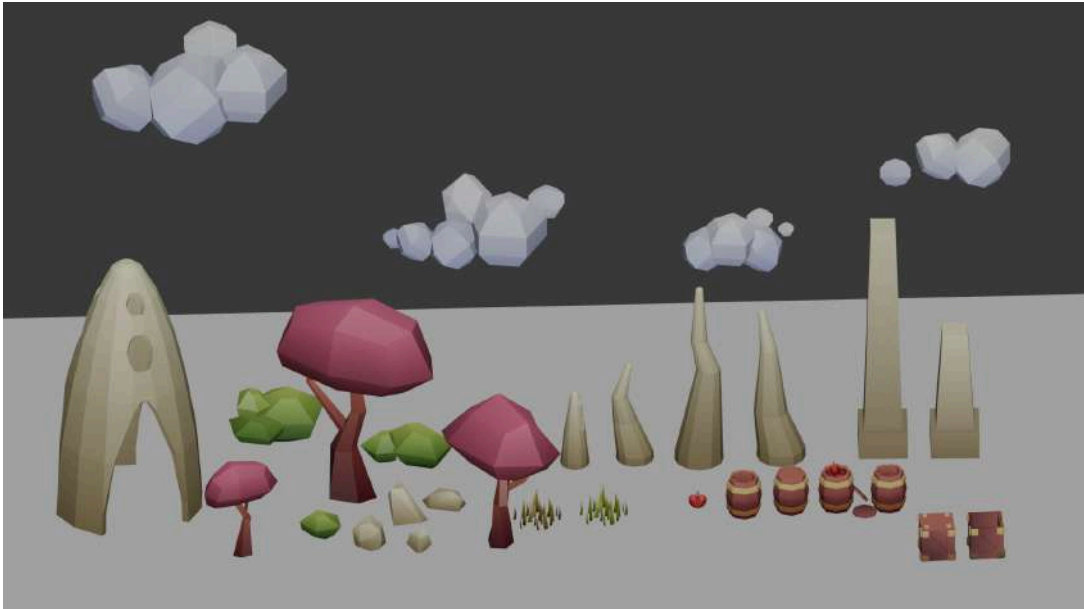


Figure 31: Final 3D models of the environment props.

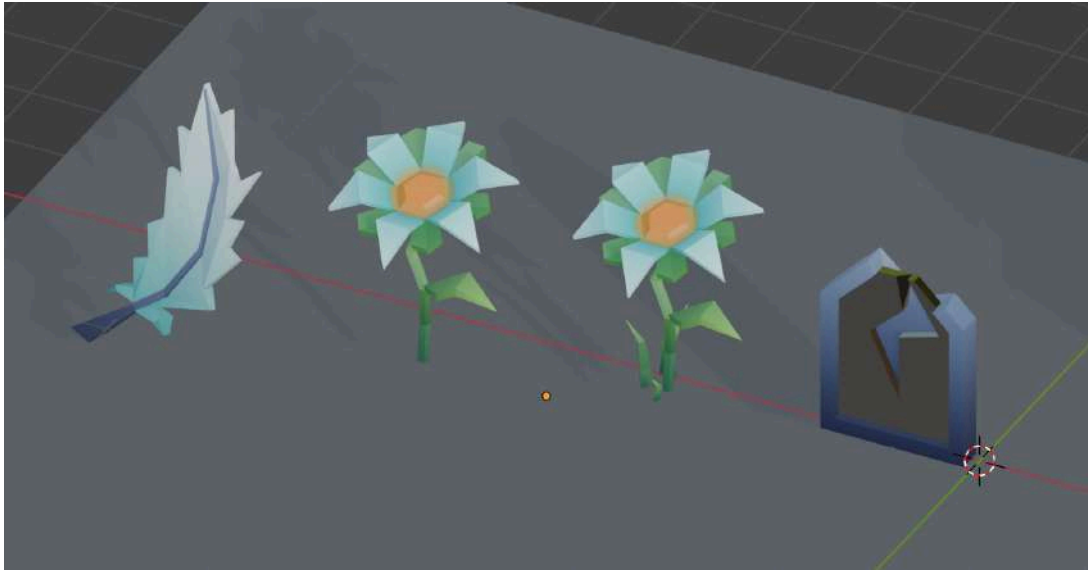


Figure 12: Screenshot of the objects needed to get the Artifact (see section 4.7).

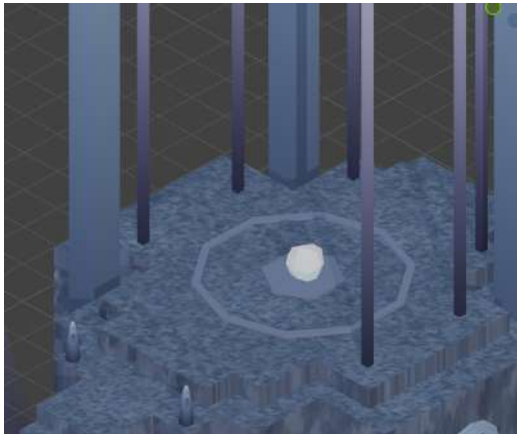


Figure 32: Final 3D model of the Sanctuary's big and thin columns.



Figure 33: Final 3D model of the Sanctuary's arch.

5.3.2. COLORING GRADIENTS

Level 01

Object	Lighter Code (1)	Darker Code (2)	InBetween Code
Pilares, piedras, caminos, spawn, puente	#fdfccf	#8f805c	
Arbustos	#c3e368	#1c4a19	
Hierba	#fff873	#4f701b	
Tierra isla	#fdfdc7	#d1c38d	#757071
Skybox	#dee4f0	#9892dd	
Puerta	#f0fbfc	#757071	
Nubes	#ffffff	#	

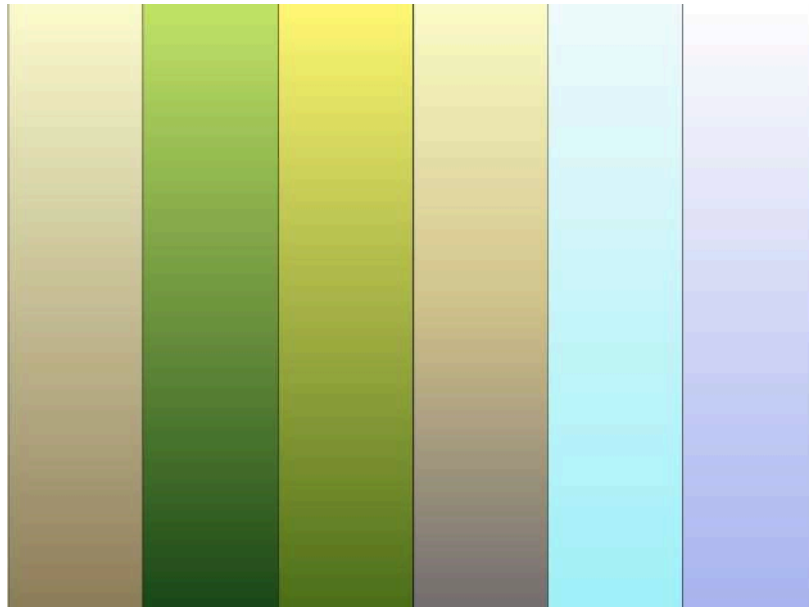


Figure 34: Coloring gradient of Level 1.

Level 02

Object	Lighter Code (1)	Darker Code (2)	InBetween Code
Castillo, pivotes, puente	#ffeb94	#995a2e	
Barriles, cajas	#cc7d6c	#52192a	
Piedra, checkpoint	#e8b3d2	#573c58	
Hierba	#ffec8a	#4f440f	
Tierra Isla	#dfa09e	#dfa09e	
Arboles (Hojas)	#f0739d	#521e27	
Arboles (Tronco)	#d98671	#4f151f	
Health Item	#ff7069	#5c1220	#ab2228

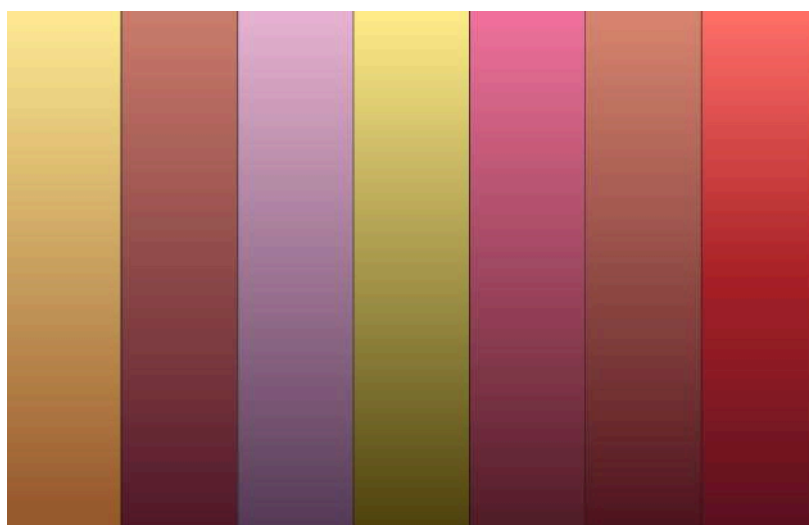


Figure 33.1 : Coloring gradient of Level 2.

Sanctuary

Objects	Lighter ode	Darker Code
Columns, Pivots,	#778eb0	#1b2040
Sword Stand	#e6e9ee	#a8f4ff
Big Rocks	#b6afe0	#1e1e39

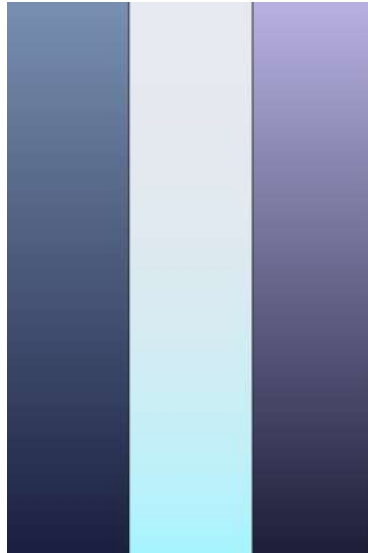


Figure 33.2: Coloring gradient of the Sanctuary

5.3.3. ANIMATION

Some assets will have small animations made with Unity. Herbs will be animated. Clouds have a simple scripts that takes them move and reappear at the beginning of axis X.

5.4 INTERFACE

The interface aesthetics are 2D and simplified, inspired by games like Zelda: Link's Awakening and Death's Door. See references in point

MENU'S LOGIC DIAGRAM

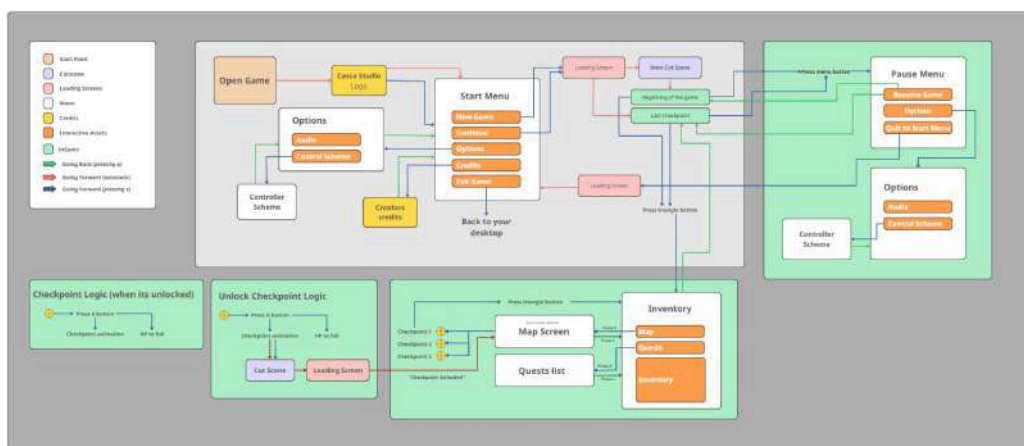


Figure 34: UASU's Menu's logic diagram.

MENU'S APPEARANCE CONCEPT

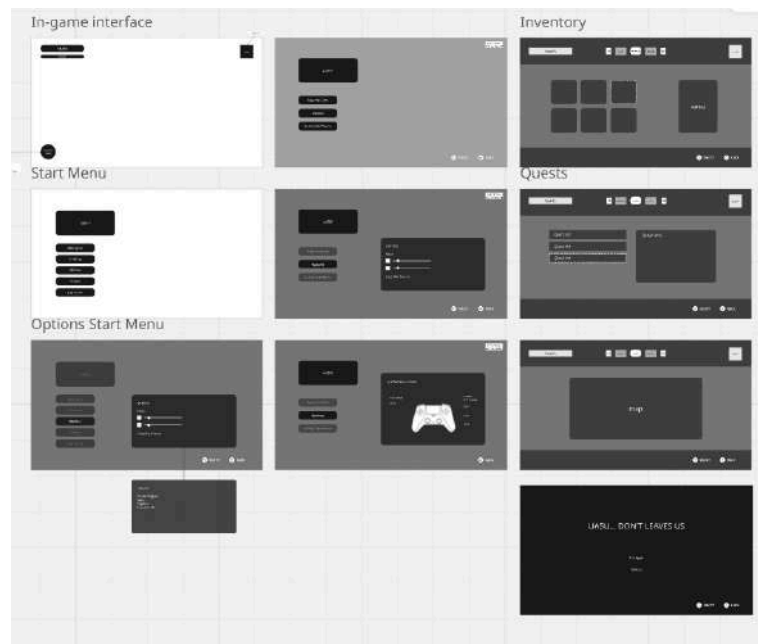


Figure 35: UASU's menu's appearance concept.

5.4.1 LOADING SCREENS

Loading screens are small animations of the protagonist in the center of the screen that provide tips to the player. They can be reminders of the game's controls or tips for progressing through missions.

5.4.2 GAME OVER SCREEN

When Uasu's health reaches 0, a black screen will appear. It will show a little message telling Uasu to not give up and 2 button options: Try Again, to continue from the last checkpoint, and give up, to exit the game.

5.4.3 START AND PAUSE MENU

The game's name will appear on the home screen, followed by the various menu options:

Start Menu	Pause Menu
New Game	Resume Game
Continue	Options • Volume • Language • Controller Scheme
Options • Volume • Language • Controller Scheme - Keyboard - Controller	Quit to Start Menu
Credits	
Exit	

For the start menu, in the background, we see a zoom in of the stage from the first part of the demo seen from a different perspective than isometric, to show the player something they haven't seen before in the game.



Figure 36: Main Menu



Figure 36.1: Options panel open.

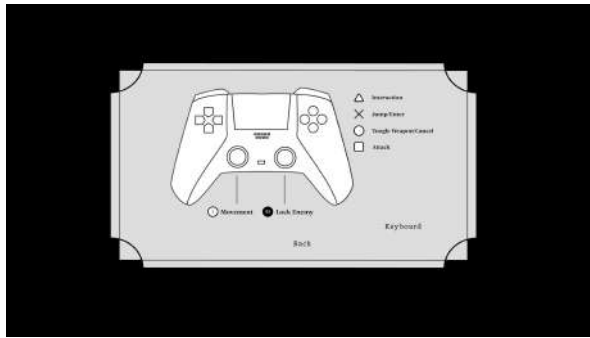


Figure 36.2: Controller Scheme panel open.

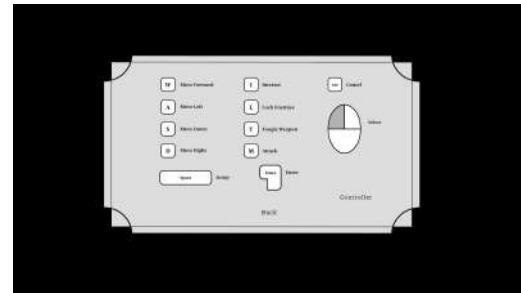


Figure 36.3: Keyboard Scheme paanel open.

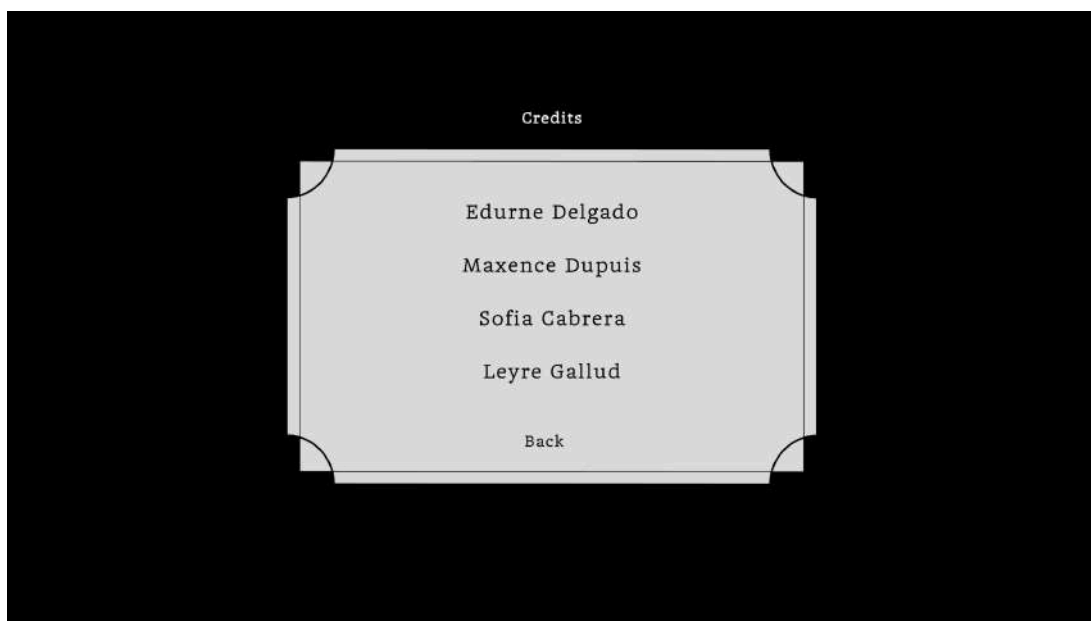


Figure 36.5: Credits panel open

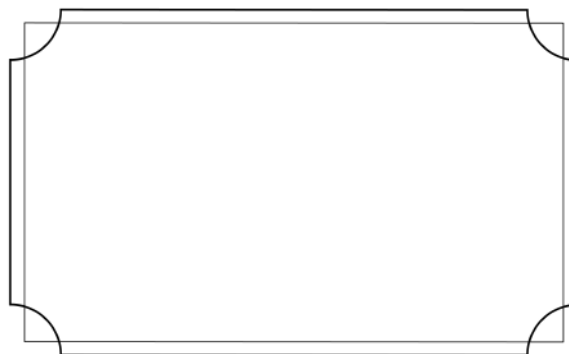


Figure 37: Menu Panels' frame box.

Figure 38: Start Menu's Buttons.

Figure 38.1: Start Menu's Buttons (highlighted).

CONTROLLER SCHEME



Figure 39: UASU's controller scheme.

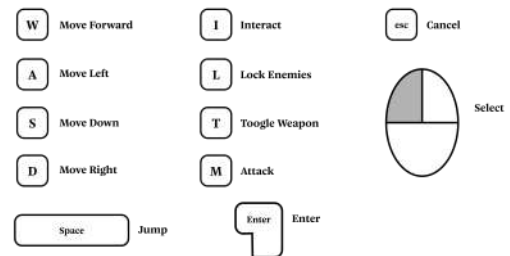


Figure 39.1: UASU's keyboard and mouse control layout.

For the pause menu, black background with opacity adjustment and the options display in the center. Follow references in [section 9](#).



Figure 40: UASU's Pause Menu.

5.4.4 LIFE BAR



Figure 41: Uasu's Life Bar.

- **Protagonist:** Reduced health bar in the upper left corner. Fixed position and visibility throughout gameplay. The healthbar changes its color with a gradient while losing HP.



Figure 41.1: Healthbar Border UI Design.

- **Enemies:** Health bar that appears above enemies when you start a fight with them.
- **Boss:** Large health bar located in the lower half of the screen. It takes up half the entire width of the screen and is decorated with a 2D border.

5.4.5 SOULS



Figure 42: Screenshot of UASU showing a Soul in the game.

There's a counter on the player's sword for the number of souls they've collected. Upon acquiring one, the counter appears in the upper right corner for 5 seconds. After that time, it disappears. Players can also check the number of souls they have collected in their inventory.



Figure 42.1: Final design of UASU's souls.

5.4.6 HELP INDICATORS



Figure 77: Beginning of the game when Uasu has to wake up

When the player must press an specific key to perform an action, and little text in white will appear on top of Uasu telling them which type of key they need to press.

INTERACT mean they must press the key that does the action of interact, info you can find in the Controller Scheme inside the Options Menu

5.4.7 INVENTORY (FEATURE NOT ADDED)

Black background with medium-low opacity, the background is blurred. The inventory has three sections:

- **Map:** It is unlocked as you activate save points. You can see the current location and checkpoints.
- **Quests**

- **Inventory:** Where the player collects items: souls, swords, talismans, etc.

5.4.8 DIALOGUES

Dialogues with other characters appear in a rectangular box at the bottom of the screen with some 2D decoration around the edges.

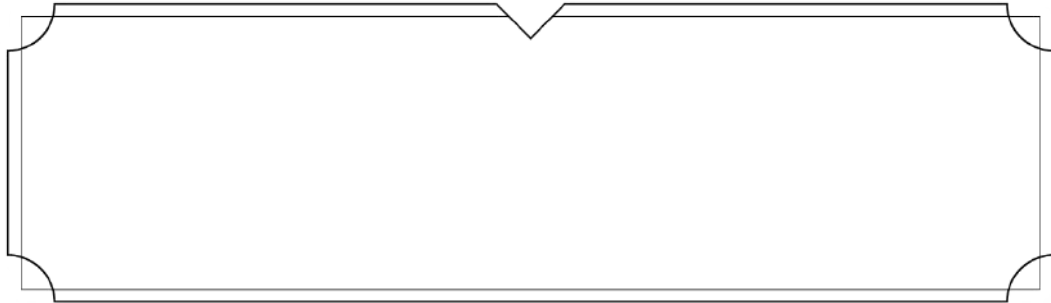


Figure 43: Final design of the dialogues panels.

Some dialogues will have options, and they will appear inside this buttons.



Figure 43.1: Final design of the options panels.

5.4.9 OTHER ICONS

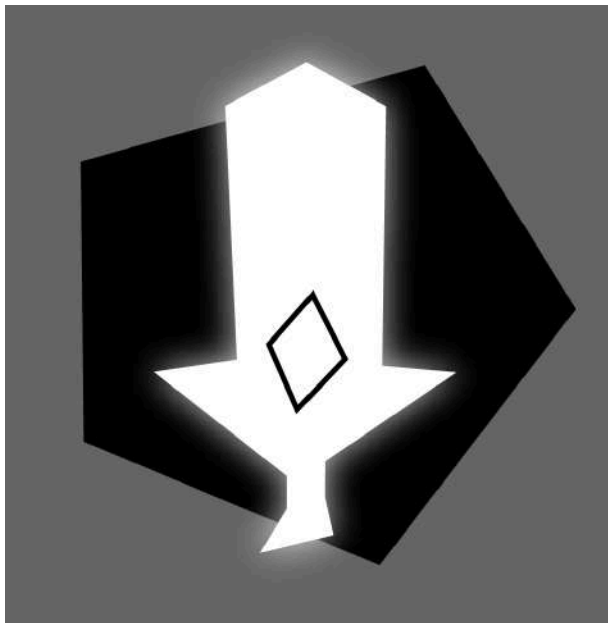


Figure 44: Final design of the Sword Icon

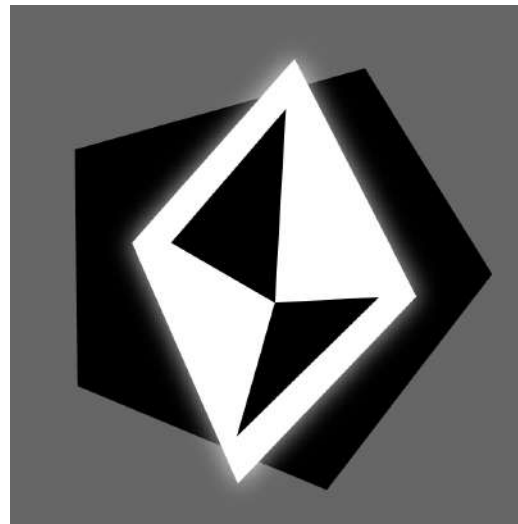


Figure 44.1: Final design of the Artifact Icon

5.4.10 FONTS

We will use 3 different fonts:

- **Main Logo.** Dottera for the game's logo.

The logo for UASU, featuring the letters 'U', 'A', 'S', and 'U' in a stylized, bold, sans-serif font. The 'A' is unique, with a dot on its left vertical stroke.

Figure 45: UASU's main logo.

- **Prophecy.** Heartbeat Synchronicity for the intro cinematic.

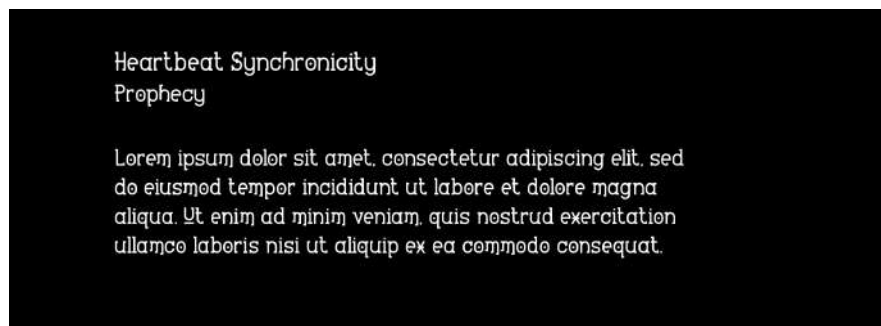


Figure 46: Typography used for the introductory cinematic.

- **Text.** Nice VAR for the rest.

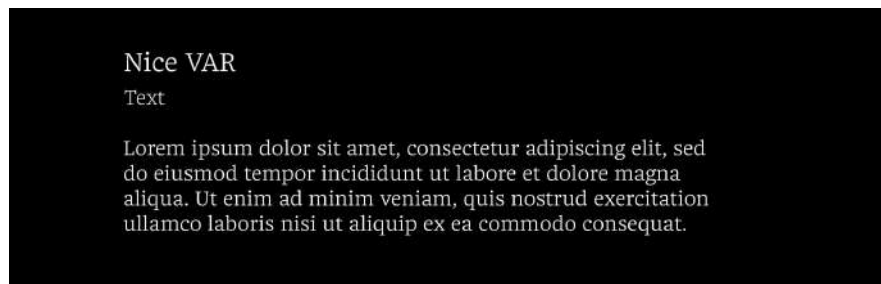


Figure 46.1: Typography used in UASU's texts.

5.5 PARTICLES

In our game there are two types of particles: ones that are constant and are always playing, and those which are activated by the players interaction.

Visually there are also two types, leaves of different colors that fall from bushes and trees, and sparkles that are on top of sculptures and pivots, and that appear when you interact with healing items and the Checkpoint Statue.



Figure 78.1 Checkpoint Statue with particles

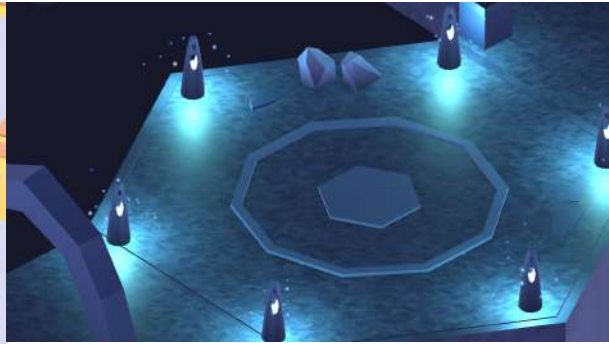


Figure 78.2 Sanctuary Pivots with particles

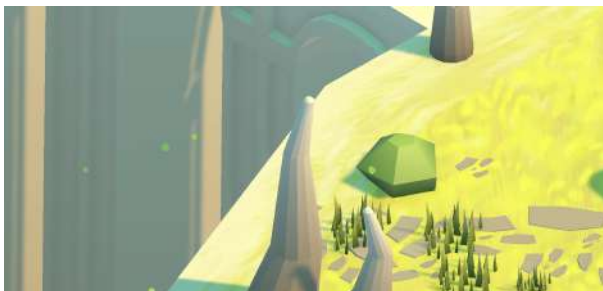


Figure 78.3 Bushes in level 01 with particles

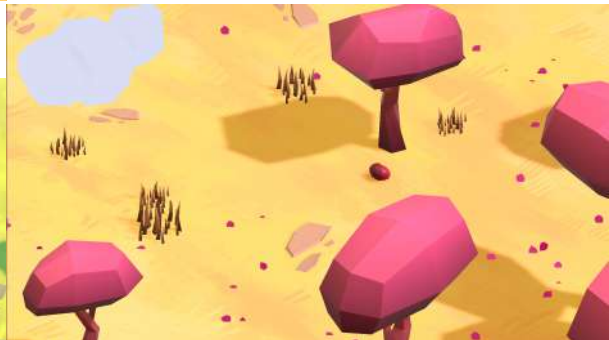


Figure 78.4 Trees in level 02 with particles

5.6 AUDIO SYSTEM

5.6.1 MUSIC

With the game's music, we build an atmosphere that evokes adventure and fantasy without intense rhythms, but rather contemplative and mysterious ones; using some of the following instruments complemented with Dorian mode scales or Mixolydian:

- Marimba
- Harp-Flutes
- Glockenspiel or celesta (small metal plates) (*fantasy*)
- Pizzicato, a family of bowed strings (*curiosity*)

Examples: Tales of Symphonia (Deepest Woods), Gormotti Forest (Xenoblade Chronicles 2), Secret of the Forest (Chrono Trigger)

The most appropriate music genre would be symphonic. Instrumental music divided into two main themes:

- The first plays outdoors, i.e., levels 1 and 2. This theme is lively, with a rhythm that invites the player to explore. The sanctuary will feature the same song but with a slower tempo and a different scale to create a sense of mystery and fantasy.- Inside, we have a darker and more eerie song.

5.6.2 SOUND EFFECTS

The game places importance on sound effects. They are implemented in most of the characters' actions: when they walk, when they interact, when they hit, etc. Likewise, sound effects are included when interacting with the interface itself.

It is worth noting the characteristic Siffler whistles, in the protagonist's dialogue, and in enemy interactions. The latter feature a sound modification/deformation to convey to the player that the enemies are Siffles, but have undergone some kind of mutation.

Action	Type
Walking	InGame,Uasu
Sword Swing	InGame,Enemies
Whistle for the Checkpoint	InGame
Enemies Whistle	InGame,Fighting,Enemies
Enemies Dying	InGame,Fighting,Enemies
Selection Sound	UI
Click Sound	UI
Enemies Hitting You	InGame,Fighting,Enemies
Sorcerer Voice	InGame
Uasu Voice	InGame
Jumping Sound	InGame,Uasu
Getting Item	InGame
Healing	InGame
Getting Artifact	InGame, Dialogue

5.6.3 MUSIC REPOSITORY

Place	Song	Links
Start Menu	Moss Grotto. <i>Silksong</i>	https://spotify.link/fANV5NmwwXb
Level 1	Gormotti Forest. <i>Xenoblade Chronicles 2 OST</i>	https://spotify.link/5AyhVcxvwXb
Level 2 (pt. 1&2)	Secret of the forest. <i>Chrono Trigger OST</i>	https://spotify.link/RNXwbgzvwXb
Level 2.3	Deepest Woods. <i>Tales of Symphonia</i>	https://spotify.link/aotA2cyvwXb
Level 3	Cave music quest. <i>The Legend of Zelda</i> . Nintendoland.	https://youtu.be/LowOWwgJ1V0?si=O4bCkV7XHydC3huK
Level 4 (entry)	Silence? Distorsion and whistle	
Level 4 boss	Mantis Lords. <i>Hollow Knight</i>	https://spotify.link/4J25hTlwwXb
Intro Cutscene(prophecy)	Wind Scene. <i>Chrono Trigger OST Hill</i>	https://spotify.link/zR3jlJAvwXb
Ending Cutscene	<i>The Ballad of Twilight. The Legend of Zelda Twilight Princess</i>	https://youtu.be/a8p8SVQ8gKQ?si=YjMtrM0t1qywMMAY

6. STORY & NARRATIVE

6.1 BACKSTORY

In a vast world where many races coexist there's one called "The Sifflers". They are a pacific race that was enslaved for many years, exploited by warlords because of their shyness, harmless appearance and hardworking behaviour. They worked in the mines, the fields, they build weapons, etc. By many races, they were considered "inferior", not even living creatures, so they didn't even have a name.

After the wars, they gradually and peacefully gained independence and freed themselves from the slavery—yet remained victims of prejudice by some people. But things were changing. Slowly, the Sifflers built their place in society, they started to get recognised as an official race and not just as working machines, so this is when they received the name of "Sifflers", because of the whistling sound they used to communicate. They developed unique skills and crafts thanks to their ancestors labor, and the whistling language they created during war times was established as their official language.

Sifflers were finally able to live peacefully, but one day they started to disappear. Their villages were getting emptier day by day and no one knew what was happening.

Who was behind it, and why?

6.2 MAIN PLOT

The game follows the prophecy of Uasu, the Chosen Siffler, who is given the mission to save his race by freeing them from the control of a vengeful spirit.

The plot is linear timewise. Players will see the story develop from start to finish without time skips. However, backstory and context for certain things will be explained only through dialogue with npcs, some of it being optional and skipable.

The first thing we will see when we start the game is Uasu waking up on a field of flowers, a place that is not familiar to him. After whistles from an unknown source start speaking to him, telling him that "he must save them all", he will start to investigate the mysterious island. A big castle rises in the horizon, and Uasu's journey seems to be in its direction. And the first obstacle to get to it is a big stone wall.

After he gets through, the entrance of the building can be seen, but various mysterious monsters guard it. Weak and weaponless, Uasu tries to find a way to face them by entering the Wild Forest, when a strange individual hidden by some trees calls out to him. "The Sorcerer", he says he's called, and in spite of his suspicious and weird appearance, he will provide him with very useful information. Places he could go, things he could need for survival, and most importantly: That those monsters are Sifflers transformed, and that something inside the castle is behind it.

Looking through the forest, Uasu will find an old closed off sanctuary behind some bushes, which requires an offering to open. An offering related to an Artifact he has to get from the Sorcerer. After getting something to trade the artifact for, he heads to the sanctuary again and acquires a Magic Sword he will have to use to fight monsters and save them from the spell that controls them.

Freeing the guards by the entrance and getting all the souls required, he gets into the castle lobby. To get to the king's chamber he will have to endure some puzzles and attacks from monster Sifflers along the way, and when he finally reaches the end of the hallway, he will have to unlock a portal by using a key he found along the way. Getting through it, he found himself in a dimly lit room filled with cages, war weapons and Siffler Statues. In front of it, a big ghostly figure. And that figure kinda looks like him.

We soon realize that it is an ancient and vengeful ghost of an old Siffler, telling Uasu that they are creating an army of strong Sifflers to get revenge from all those years of slavery in the past. The Ancient Siffler and Uasu enter a fight, having to also fight some of the transformed Sifflers the ghost is using against him. Uasu ends up winning, but to free and transform back all the monster Sifflers, he has to sacrifice something: his own soul. Which he does.

6.3 CUTSCENES

They will be done in a complementary 2D artstyle that contrasts the 3D of the rest of the game. They are "Pieces of the Prophecy", similar to games like Breath of the Wild, and will be shown as short animatics.

In this Demo, there will be an introduction cutscene that will show the backstory of the Siffler race, and an ending cutscene, which will leave the ending open with a hint of a battle that could be implemented in the final version of the game.

It is done using illustrations made in procreate and adobe after effects for the little animations and text.

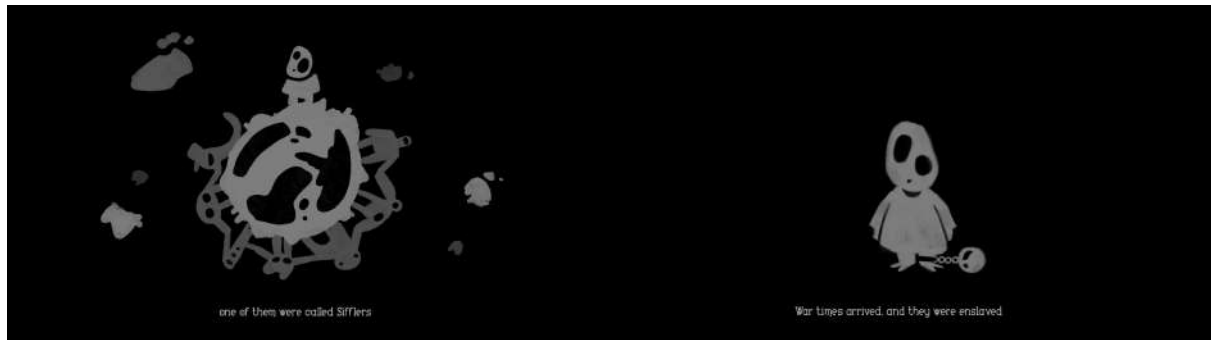


Figure 47: Frame from the introductory cinematic.

Figure 47.1: Frame from the introductory cinematic.



Figure 47.2: Frame from the introductory cinematic.

Figure 47.3: Frame from the introductory cinematic.

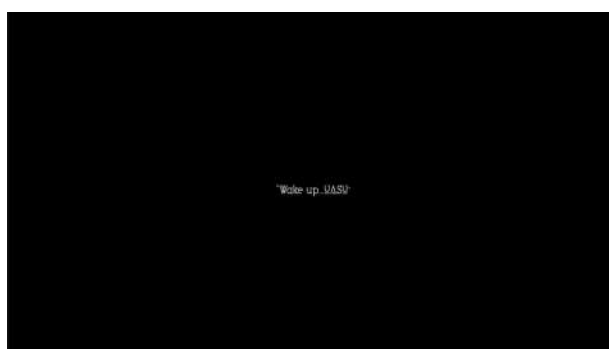


Figure 47.4: Final frame from the introductory cinematic.

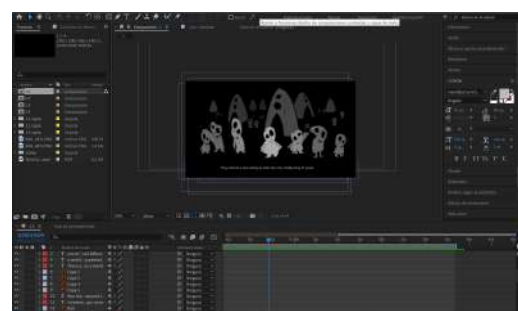


Figure 47.5: Introductory cinematic sequence shown within the animation/compositing workflow during development.

7. CHARACTERS

7.1 MAIN CHARACTER

Uasu, a normal Siffler turned into a hero. He is inspired by Shy Guy (Mario Bros Franchise), the Hollow Knight, the Kodama creatures from Princess Mononoke and Link in Links Awakening. He is from a fictional race called Sifflers, name that comes from the french word Siffleurs, which means "Whistlers". They receive this name because of their way of speech, which is by making whistling sounds. The Siffilent, or Whistle, is a secret language they created when they were enslaved so their owners would not understand what they talked about. It was passed onto the next generations, until it was the official and only language they speaked. He is a vessel through which we tell the story, so he doesn't have a very outstanding nor interesting personality. He is just very chill individual who accepts his destiny, motivated by his connection with his people, by whom he is guided. He represents the Sifflers of the present. A generation that has forgiven and moved on from its past, being grateful to their ancestors for what they went through but accepting that revenge wouldn't go anywhere.

7.1.1 APPEARANCE

His general appearance is cute, but cool. Small, simetric, soft and round body proportions. A mask like face with two asymmetric black holes as the eyes and a small mouth, features all Sifflers share. As clothes, he wears a cloak with a hood, boots and a protective scarf.



Figure 48: Screenshot of Uasu in-game.



Figure 48.1: Uasu's 3D model.

7.1.2 ABILITIES

He can jump, run, swing the sword and craft potions and other types of items, with the help of a crafting station. Also, the Artifact it possesses prevents him from being hypnotized and transformed by the Ancient Siffler.

7.1.3 RELATIONSHIPS

Along the way he befriends The Sorcerer, who helps him in many ways. But he mainly stays alone on his journey.

7.2 ANTAGONIST

The Ancient Siffler, an old, huge and deformed being filled with rage and a great desire to get revenge on those who mistreated their race in the past, comes back to life as a ghostly figure that can only be defeated with the magic sword of Souls.

It's the reason why all the Sifflers started disappearing, since its main objective is to lure all of them into it's castle and create with them an army of monsters that will help him get revenge.

It represents the oldest generation of enslaved Siffers, and its the final boss you will have to fight to "finish" the game. The corrupted Siffers will act as enemies you will have to fight to get through the levels.

Their appearance, sharp, irregular and dark, contrast with the main characters' colorful and soft silhouette. There will be different evolutions of this enemies:

- 1 grade that you can find roaming the map, make small damage.



Figure 49: Screenshot of the Grade 1 Corrupted Siffler in-game.



Figure 49.1: Grade 1 Corrupted Siffler's 3D model.

- 2 grade that are several siffers combined, making more damage.

7.3 SUPPORTING CHARACTERS

The Sorcerer/Wizard is the only secondary character you will interact with. He will give the protagonist information about the world, history, the Siffler race, tricks and advice on how to progress the game and, most importantly, the Artifact to get the Magic sword.

He will talk in riddles, explaining everything in a very ambiguous and mystical way so the player has to put effort into connecting the dots and making everything make sense.



Figure 50: Screenshot of the Sorcerer in-game.

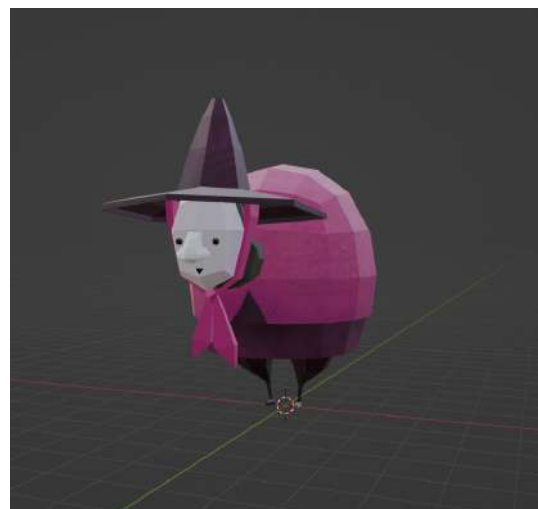


Figure 50.1: Sorcerer's 3D model.

8. CONCEPT ART

8.1 UASU

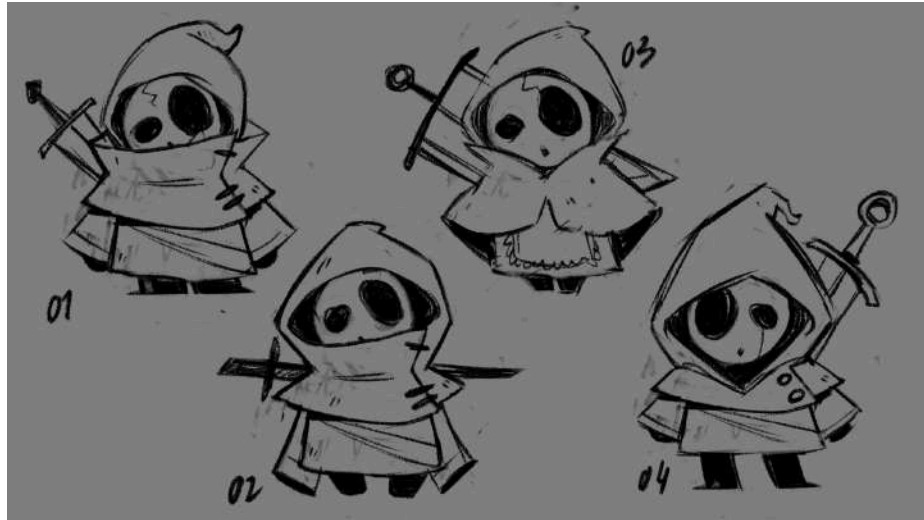


Figure 51: Early Uasu concept sketches.



Figure 51.1: UASU's final design: Action and Expression Studies

8.2 SORCERER



Figure 52: Concept art of the sorcerer: two early design iterations of the sorcerer character (see section 7.3 for the final design).

8.3 BOSS

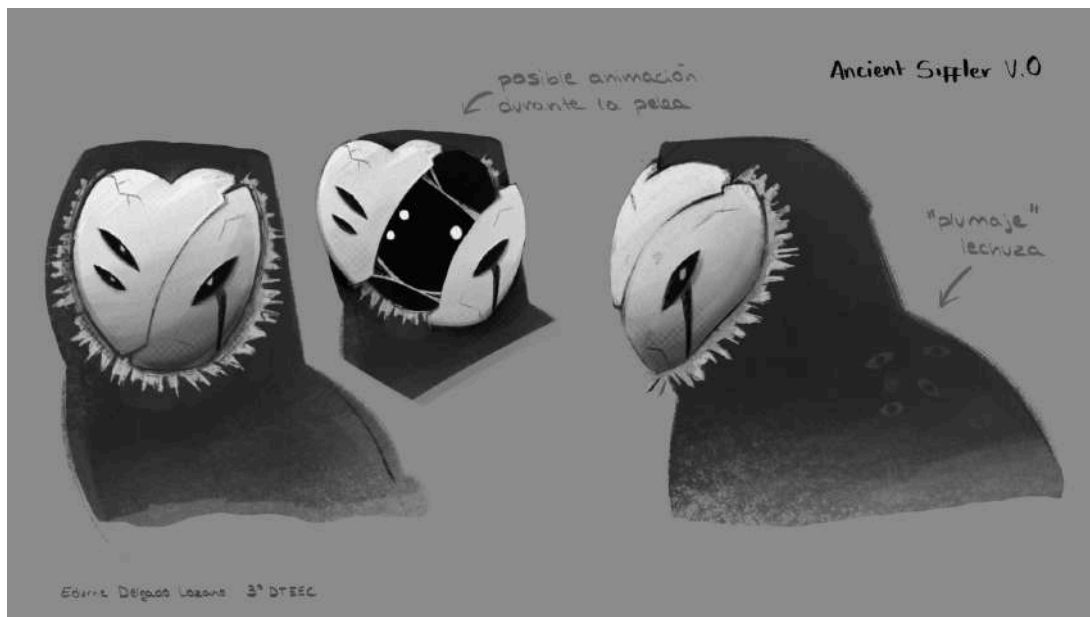


Figure 53: Concept art of the Ancient Siffler.

8.4 ENEMIES



Figure 54: Concept art of the monster sifflers.

8.5 INTERACTABLES

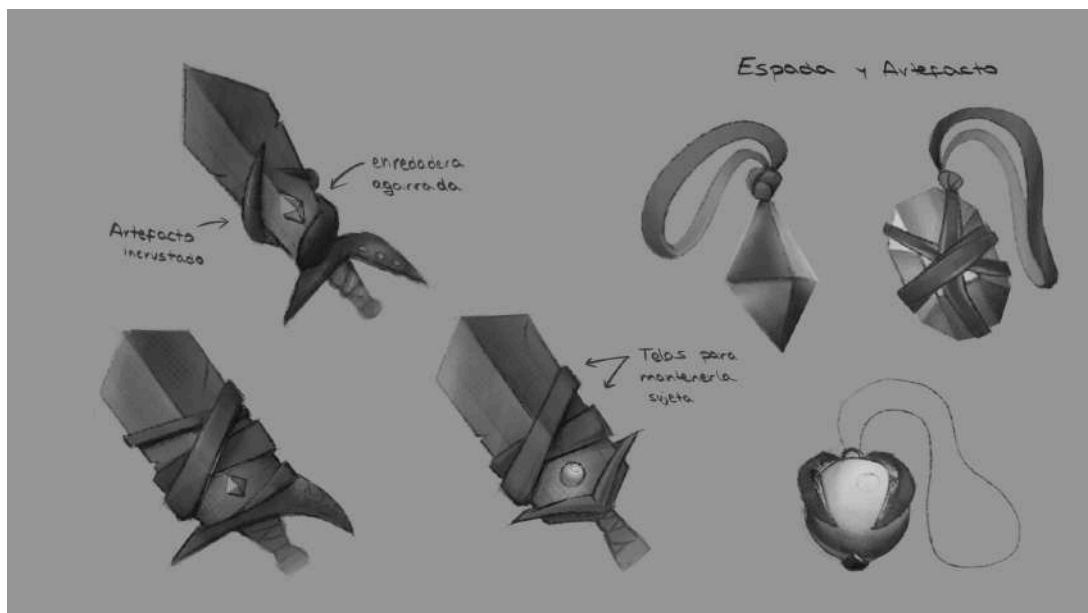


Figure 55: Concept art of the sword and the Artifact: early design iterations of the sword and Artifact (see section 10.4 for the sword's final design).

9. REFERENCES

9.1 INTERFACE

9.1.1 INVENTORY



Figure 56: Screenshot of *Death's Door* inventory.



Figure 56.1: Screenshot of *Death's Door* inventory.

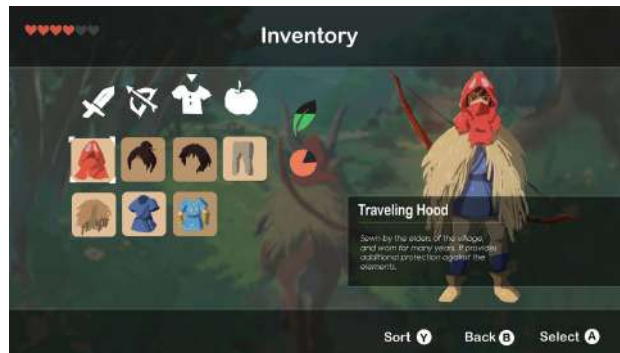


Figure 56.2: *The Legend of Zelda: Breath of the Wild* 's inventory combined with elements from the movie *Princess Mononoke*.



Figure 56.3: *The Legend of Zelda: Breath of the Wild* 's inventory.

9.1.2. START MENU



Figure 57: *Rime* 's Start Menu.

9.1.4. OPTIONS POP-UP

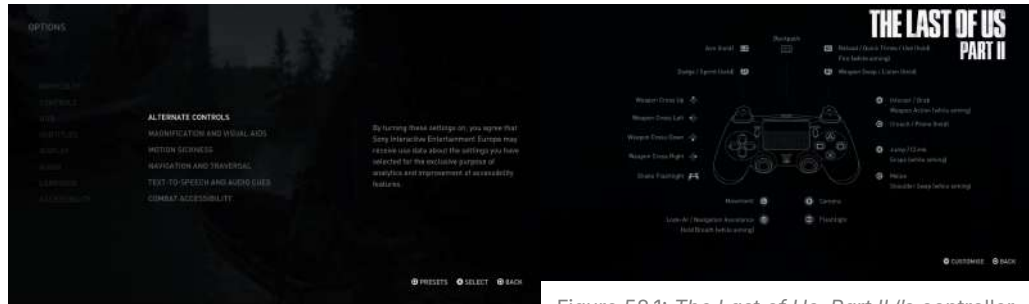


Figure 58: *The Last of Us: Part II* 's Options Accessibility Menu.

Figure 58.1: *The Last of Us: Part II* 's controller scheme.

9.1.5. LIFE BARS



Figure 59: Screenshot of *Death's Door* gameplay.



Figure 59: Screenshot of *Blasphemous'* gameplay.

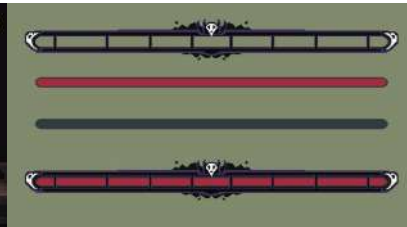


Figure 60: Final Boss Health Bar from *The Druid* by artist

9.2 CHARACTERS



Figure 61: Shy Guy, character from the *Super Mario* franchise.



Figure 61.1: Toon Link, a version of the main character from the *The Legend of Zelda* franchise.



Figure 61.2: Another version of Link, the main character from the *The Legend of Zelda* franchise.



Figure 61.3: A crow used as visual inspiration for the Monster Sifflers' design.



Figure 61.4: An owl, used as visual inspiration for the Ancient Siffler's design.



Figure 61.5: A House Sparrow, used as visual inspiration for Uasu's design.



Figure 61.6: Kodamas tree spirits from the *Princess Mononoke* movie, visual inspiration for the Sifflers' design.

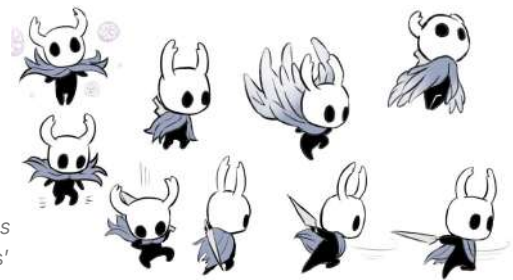


Figure 61.7: The Knight from *Hollow Knight*, used as inspiration for Uasu's design.

9.3 LANDSCAPES

9.3.1. OUTSKIRTS



Figure 62: Illustration by Oswaldo Kato. Kato's art inspired the environment's look for the game.



Figure 62.1: Illustration by Oswaldo Kato.



Figure 62.2: Illustration by Oswaldo Kato.



Figure 62.3: Illustration by
Oswaldo Kato.

9.3.2. COURTYARD



Figure 63: Illustration by Nathan Dupoy.



Figure 63.2: Landscape by Dmytro Dudko for
the game *Eldest Souls*.



Figure 63.3: Gameplay from *Cult of the Lamb*.

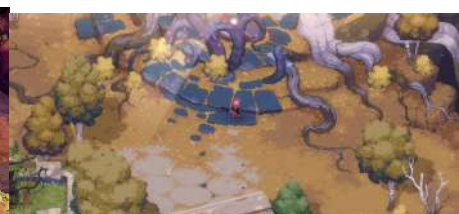


Figure 63.4: Landscape by Dmytro Dudko for
the game *Eldest Souls*.



Figure 63.5: Landscape by Dmytro Dudko for
the game *Eldest Souls*.

9.3.3. SANCTUARY



Figure 64: Cave from Death's Door.



Figure 64.1: "Sacred Seasons" game concept by Alex Pushilin.

9.3.4. CASTLE



Figure 65: Concept for the game EqqO.



Figure 65.1: Screenshot from the game 1428: Shadows over Silesia.

9.3.5. BOSS LAIR



Figure: 66: Illustration by Justin Sweet.



Figure 66.2: environment by Camille Peyreberre.



Figure 66.1: Scene from the game Eldest Souls.

9.4 MUSIC



Figure 67: Tales of Symphonia
<https://open.spotify.com/intl-es/album/6TiS9kMfpqiUeau5lX7VY2?si=pYXZ9S9rRUS9VZ-EAdVQ-w>



Figure 67.1: Chrono Trigger
<https://open.spotify.com/intl-es/album/2kOD9QzcUkeT57RjKvNvQpH?si=jvUB8TEETESg0as-W7ILWg>



Figure 67.2: Undertale
<https://open.spotify.com/intl-es/album/2M2Ae2SvZe3fmzUtlVOV5Z?si=T44RFaNXQNWz0ifRjDc5g>

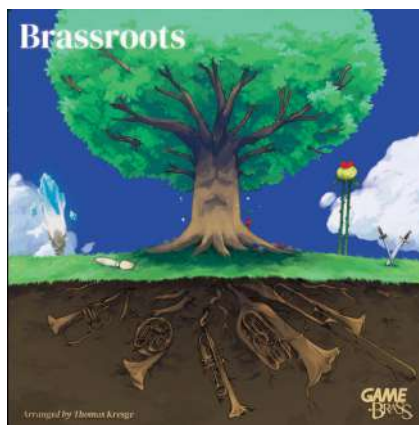


Figure 67.3: Xenoblade Chronicles 2
<https://open.spotify.com/intl-es/track/3sM2sdbKgWMMUKRw84QGdn?si=fd094fdae49c4e25>



Figure 67.4: The legend of Zelda
<https://open.spotify.com/intl-es/album/5zFkQHvRimPKxjwDwkkeNL?si=vQwC7hpHSVWHIK3Q1Bdgxw>



Figure 67.5: Silksong
<https://open.spotify.com/intl-es/album/2mvEK1s3lpArLiUVRkqoD5?si=om6tuReeSP-NHehlvjXW0Q>



Figure 67.6: The legend of Zelda:
 Twilight Princess
<https://youtu.be/a8p8SVQ8qKQ?si=hygx-EbLT60R6MEN>

9.5 CUTSCENES

9.5.1. PROPHECY



Figure 68: Legend of Zelda: Breath of the Wild.



Figure 68.1: Deltarune.



Figure 68.2: Legend of Zelda: Tears of the Kingdom



Figure 68.3: Deltarune.

10. 3D

10.1 UASU



Figure 69: 3D Model of Uasu.

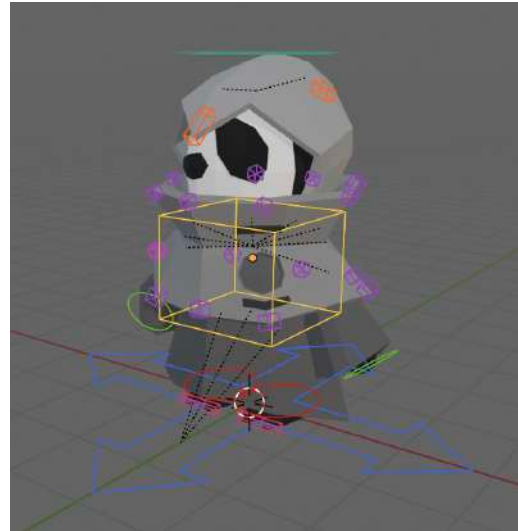


Figure 69.1: 3D model of Uasu within the animation workflow, showing the rigging.

10.2 MONSTER SIFFLER #01



Figure 70: 3D model of a Monster Siffler.

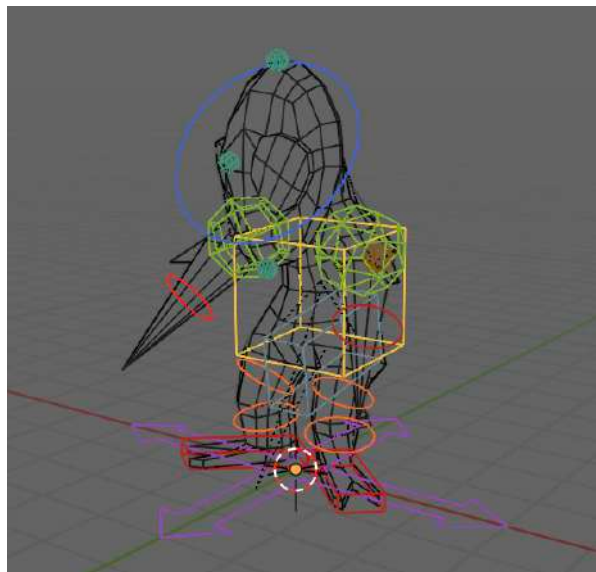


Figure 70.1: 3D model of aa Monster Siffler within the animation workflow, showing the rigging

10.3 SORCERER



Figure 71: 3D model of the Sorcerer.

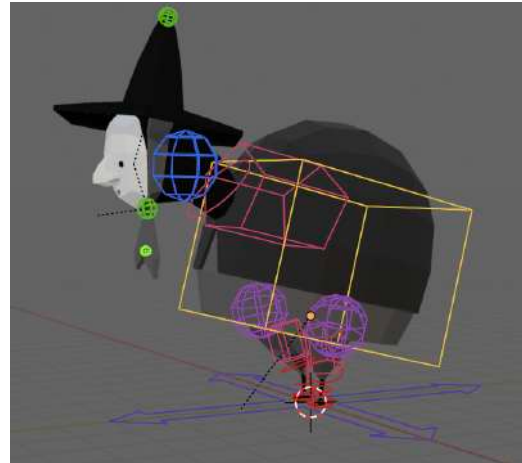


Figure 71.1: 3D model of the Sorcerer withing the animation workflow, showing the rigging.

10.4 SWORD

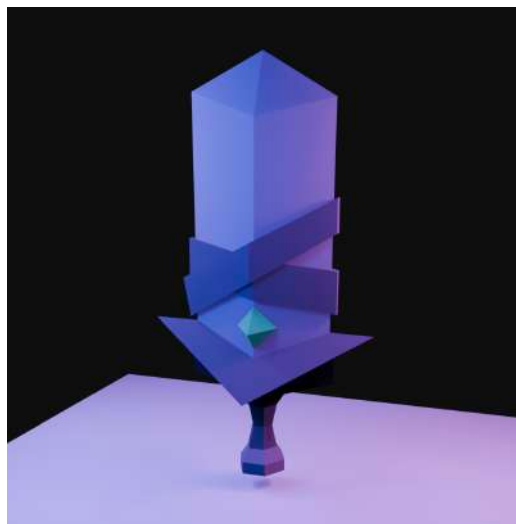


Figure 72: 3D model of the sword.

10.5 ANCIENT SIFFLER - BOSS (SIMPLIFIED VERSION)

The boss does not appear in the demo. However, we created a 3D model of his face to get a first look and for possible future iterations of the game.



Figure 73: 3D Model of the boss face

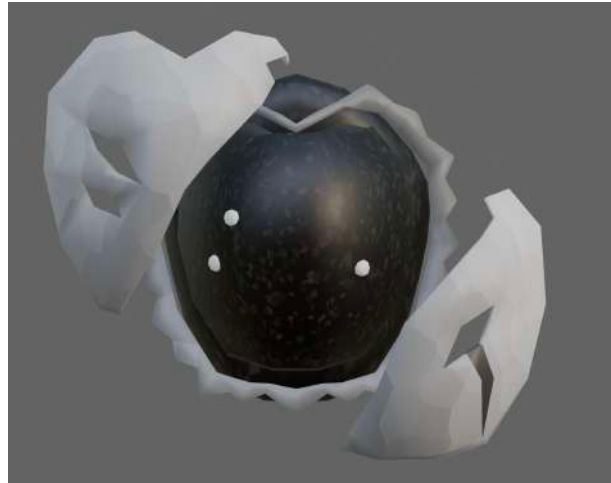


Figure 74: 3D Model of the boss face (vulnerable)

11. CODE



We aim to always get a clean, scalable and dynamic code.

11.1 GIT

Git is a powerful versioning tool that enable us to save and share our code data, create branches to work in a specific space without compromising other's stuff.



Git is not Github/Gitlab.

GitHub is a server that uses Git to enable users to easily share code through repositories and collaborate.

We use Github to share our code into a remote repository called Uasu.

Link → <https://github.com/smallboyyc/Uasu>

We have a <main> branch that represents the **"always playable version of our game"**. No one should and no one can make edition directly on <main>.



Workflow

We established a solid workflow to work with all along the project. Which is (basically):

1. **As a developer/tech artist, I want to create a new feature for the game.**
→ Create a new branch called feat/<my-feature>
2. **I want to save my work locally when I finished a task on the feature.**
→ Add, Commit
3. **I want to enable users to access my work.**
→ Push
4. **How do I access the latest edits from <main> branch ?**
→ 2. (take care you saved your files)
→ Fetch everything on remote
→ Rebase your feat/<my-feature> branch on <main>
5. **I want to merge my feature into the main project branch.**
→ Create a Pull Request from Github
→ Wait for a revision (at least one other dev)
→ Merge

11.2 CODE

We're not going into the detail of the code because it would take a week, but let's talk about the "code vision and organisation".

We use 2 Design Patterns:

- Singleton
- State

11.2.1 SINGLETON PATTERN

Singleton pattern is used when we only want to deal with one single instance of an object in the game. If a second one is created, it will be deleted automatically. A Singleton is always easy to catch and work with all along the working directory.

Ex : Player (for us it's a single player game), Audio Manager, Dialogue Manager, Isometric Camera

11.2.2 STATE PATTERN

State pattern enable us to have more control into different states at runtime for a character or basically a system. Without a state pattern, we can have a more dense code which is harder to read and debug. State pattern is a clean way to exposes states and minimise the code running in real-time.

To implement a state pattern, we need a State Machine which is basically "the boss" and expose methods to decide what state is going to be initialised and pass to an other state.

```

public class StateMachine
{
    9 references
    public State CurrentState { get; private set; }

    4 references
    public void Initialize(State startingState)
    {
        CurrentState = startingState;
        CurrentState.Enter();
    }

    53 references
    public void ChangeState(State newState)
    {
        CurrentState.Exit();
        CurrentState = newState;
        CurrentState.Enter();
    }
}

```

Figure 75: C# implementation of a state machine managing state initialization and transitions.

```

public abstract class State
{
    22 references
    public virtual void Enter() { }
    24 references
    public virtual void Update() { }
    21 references
    public virtual void Exit() { }
}

```

Figure 75.2: Abstract State base class.

```

// -> Move
if (_playerManager.LocomotionManager.IsMoving)
{
    _playerManager.PlayerStateMachine.ChangeState(_playerManager.WalkState);
}

// -> Aerial
if (!_playerManager.CharacterController.IsGrounded)
{
    _playerManager.PlayerStateMachine.ChangeState(_playerManager.AerialState);
}

// -> Lock
if (_playerManager.LockManager.IsLockedOnEnemy)
{
    _playerManager.PlayerStateMachine.ChangeState(_playerManager.LockState);
}

// -> Attack
if (PlayerInputManager.Instance.AttackPressed && _playerManager.AttackManager.CanAttack)
{
    _playerManager.AttackManager.AttackCooldownCoroutine();
    _playerManager.PlayerStateMachine.ChangeState(_playerManager.AttackState);
}

```

Figure 75.3: Player state machine transition logic.

11.2.3 MANAGERS

Working with states and singletons is good to keep a strong code structures, but alone they are... Nothing.

This is why we use a lot of "managers" script in the code which is basically a big code block with all the logics of a behaviour or situation.

Ex : PlayerAttackManager, PlayerLocomotionManager, EnemyHealthManager ...



Basically : Manager implements the code logics with for **loops, conditionals, coroutines, maths** etc... And States are using the managers to switch between each other.

So, every-time you need to implement a feature inside the game :



Make sure to :

1. Create a branch with git
2. Create a separate scene when you implement your feature
3. Once you finished with your feature, create a Pull Request on GitHub
 - a. It's possible that we will need to rebase first to keep a clear history (if you have doubt : ask @Max)
4. After a team revision, your feature can be merge.
5. Congrats, you just implemented a new feature inside the game.

12 - BUILD

During gameplay, we play and test the game in its final version. It's important to note that the output in the editor may differ from the final build. Therefore, regular testing is crucial for quickly fixing bugs.

Here is the process :

1. Make sure to commit and be up to date with the version you want to work with.
2. Go in "Edit → Build Profiles".
3. Select the platform you want to build the game.
4. Click "Build And Run".
5. Make sure to place the builded game file in a different directory than the source code folder.

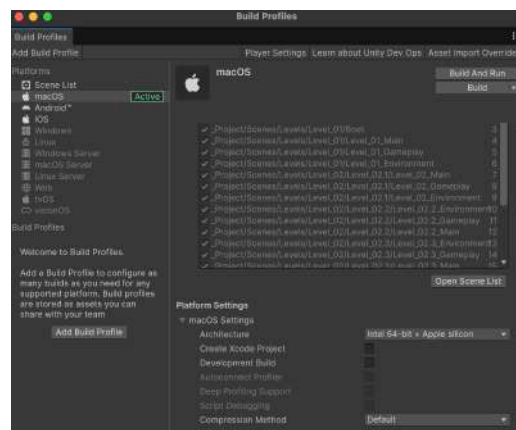


Figure 76: Build and Run game panel for macOS